IPL Mini-Project 2

Analysis of the IPL Dataset (2008-2019)

Data Science Summer Training - Batch 3

Project Snapshot

Seasons covered	12 (2008 - 2019)
Total matches	756
Total deliveries	179,078
Merged dataframe shape	179,078 rows x 39 columns
Tools	Pandas, NumPy, Matplotlib, Seaborn

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1. Executive Summary

This project analyses 12 IPL seasons (2008-2019) using ball-by-ball (deliveries.csv) and match-level (matches.csv) data sourced from Kaggle. After merging the two tables into a 179,078-row analytical dataframe and handling missing values, we engineered features for over-phase (Powerplay/Middle/Death), boundary classification and seasonal decomposition. We then explored toss impact, top performers, seasonal scoring trends, venue advantage and death-over behaviour.

The single most important insight: captains who win the toss and choose to FIELD first win 55.9% of their matches, versus only 45.7% when batting first - a consistent ~10-percentage-point edge in favour of chasing. This effect is reinforced by a clear upward trend in scoring (regression slope ~+3.6 runs/season), making defined targets easier to chase under dew on most Indian venues.

2. Data Wrangling & Feature Engineering (Part A)

2.1 Load & Merge

The two raw CSVs were merged on matches.id = deliveries.match_id using a left join, yielding a tall dataframe of 179,078 rows x 39 columns covering 12 seasons (2008-2019).

2.2 Missing Data Strategy

Missing-value counts in matches.csv: city: 7, winner: 4, player_of_match: 4, umpire1: 2, umpire2: 2, umpire3: 637.

- city -> imputed from venue (one-to-one mapping), 'Unknown' as fallback. Imputation avoids losing rows; only 7 cities were missing.
- player_of_match -> filled with 'No Award' for abandoned/tied games so the match itself is retained for win/loss analysis.
- dismissal_kind (deliveries) -> NaN replaced with 'not out' so groupby operations on dismissal types behave correctly.

2.3 Feature Engineering

- total_runs_calc = batsman_runs + extra_runs
- is_boundary -> NumPy np.where(batsman_runs >= 4)
- over_phase -> Powerplay (1-6), Middle (7-15), Death (16-20)
- year, month derived from the match date for seasonal trend analysis

2.4 Aggregations

Top 5 batsmen by total runs:

- V Kohli - 5,434 runs
- SK Raina - 5,415 runs
- RG Sharma - 4,914 runs
- DA Warner - 4,741 runs
- S Dhawan - 4,632 runs

Lowest economy-rate bowlers (>= 500 legal balls):

- A Kumble - economy 6.58
- M Muralitharan - economy 6.7
- Rashid Khan - economy 6.73
- SP Narine - economy 6.75
- DL Vettori - economy 6.78

Team with highest win % per season:

- 2008: Rajasthan Royals (13/16 = 81.2%)
- 2009: Delhi Daredevils (10/15 = 66.7%)
- 2010: Mumbai Indians (11/16 = 68.8%)
- 2011: Chennai Super Kings (11/16 = 68.8%)
- 2012: Kolkata Knight Riders (12/17 = 70.6%)
- 2013: Mumbai Indians (13/19 = 68.4%)
- 2014: Kings XI Punjab (12/17 = 70.6%)
- 2015: Mumbai Indians (10/16 = 62.5%)
- 2016: Sunrisers Hyderabad (11/17 = 64.7%)
- 2017: Mumbai Indians (12/17 = 70.6%)
- 2018: Chennai Super Kings (11/16 = 68.8%)
- 2019: Mumbai Indians (11/16 = 68.8%)

3. Key Insights with Visualizations (Part B)

3.1 Toss Impact on Match Outcome

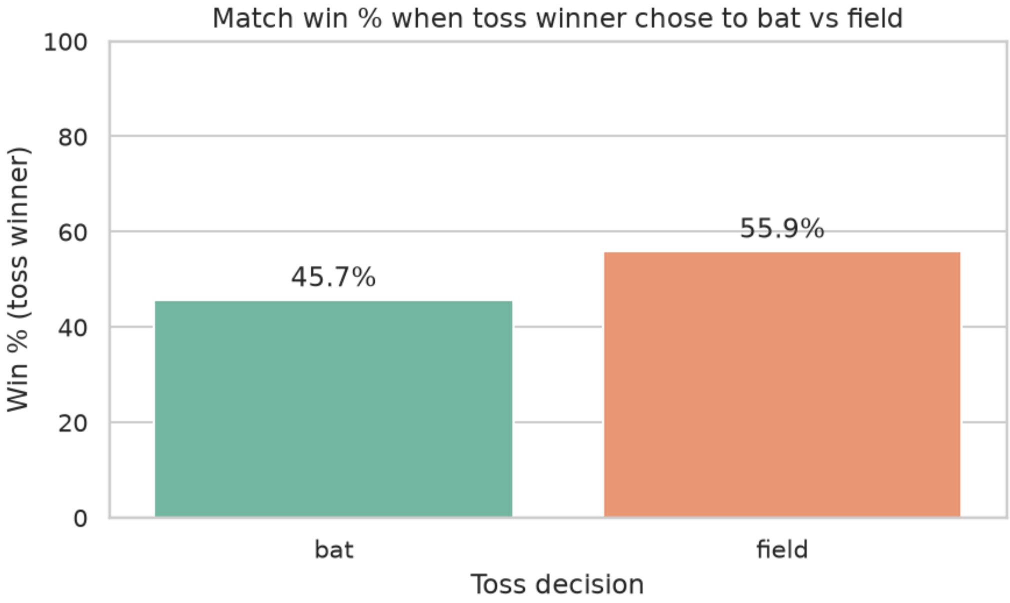

Fig 1. Win % of toss-winning teams by toss decision.

Field-first delivers 55.9% wins vs 45.7% for bat-first. The advantage holds in almost every season (see season-wise breakdown below), driven by dew, pitch slow-down in second innings and the strategic clarity of chasing.

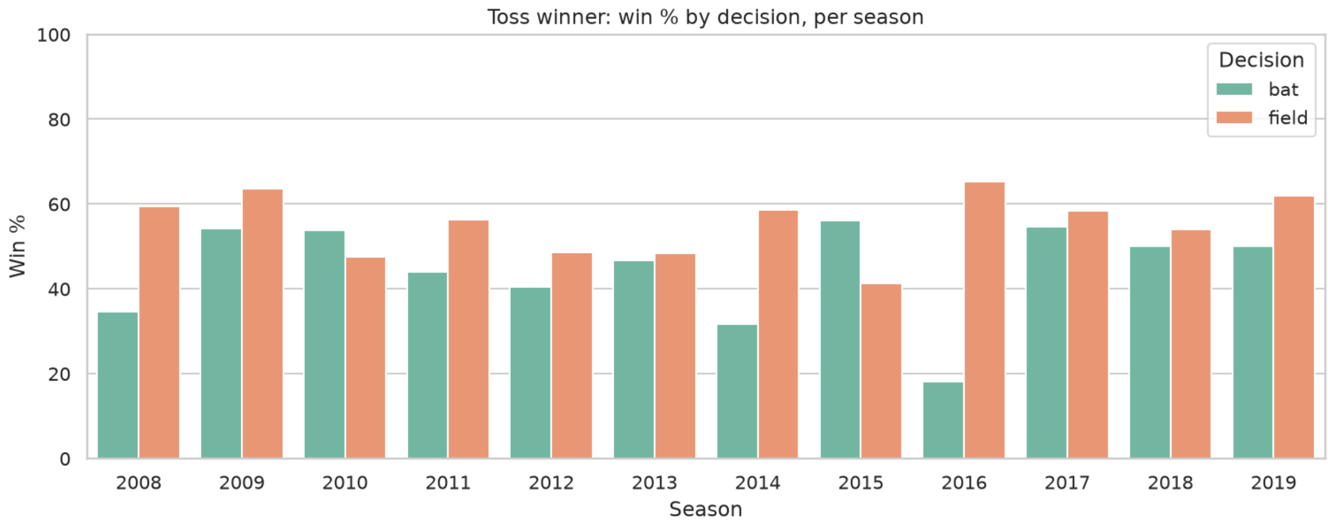

Fig 2. Season-by-season breakdown - field-first dominates after 2013.

3.2 Top Run-Scorers

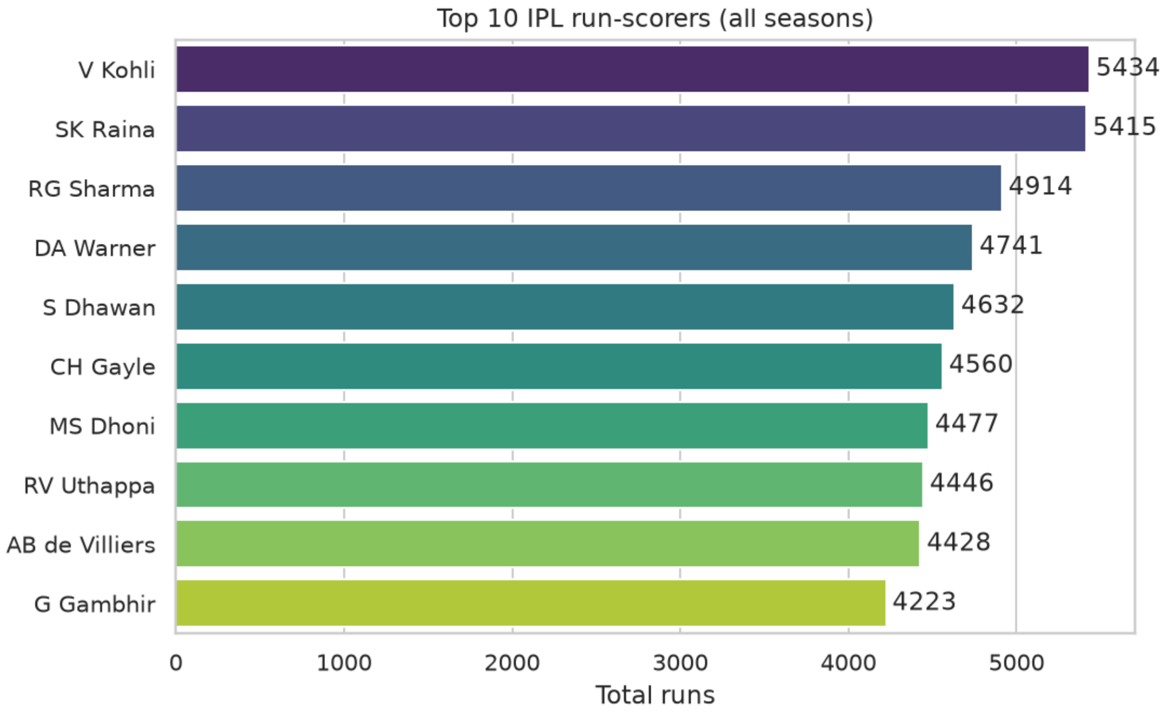

Fig 3. Top 10 IPL run-scorers across 12 seasons.

V Kohli leads with 5,434 runs, followed closely by SK Raina (5,415). The top 10 is dominated by top-order batters who routinely cash in during the powerplay - a strong argument for protecting in-form openers.

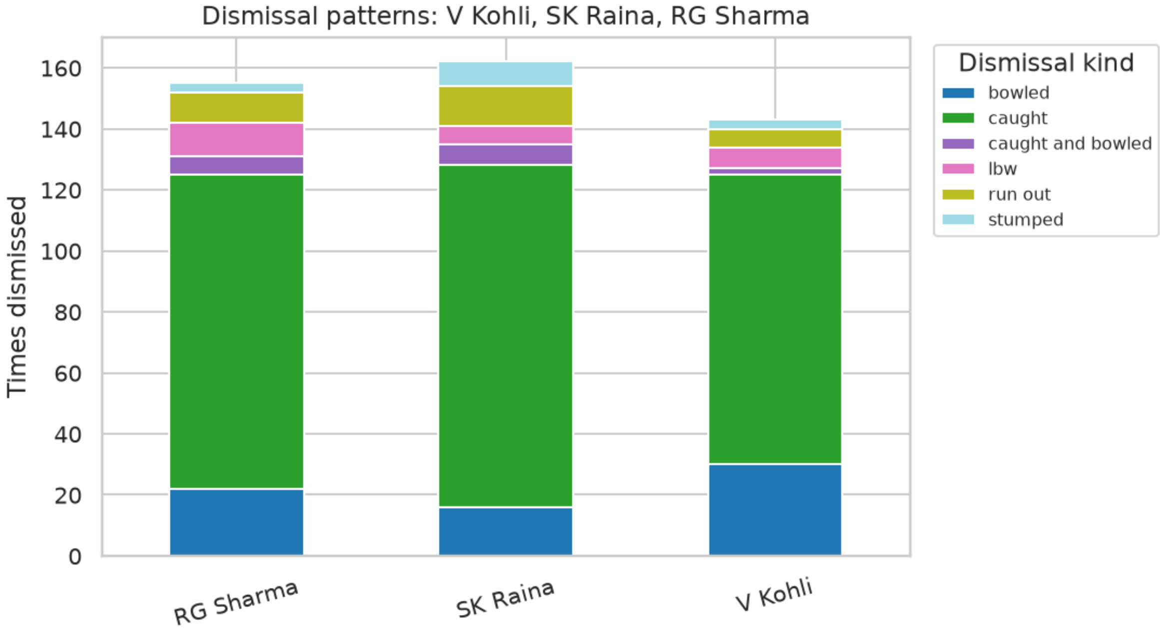

Fig 4. Dismissal patterns for the top 3 scorers.

'Caught' is the dominant dismissal mode for all three, reflecting the aerial-stroke nature of T20. Kohli and Raina also fall 'bowled' often, suggesting opposing teams target full/yorker-length deliveries against them.

3.3 Seasonal Batting Trends

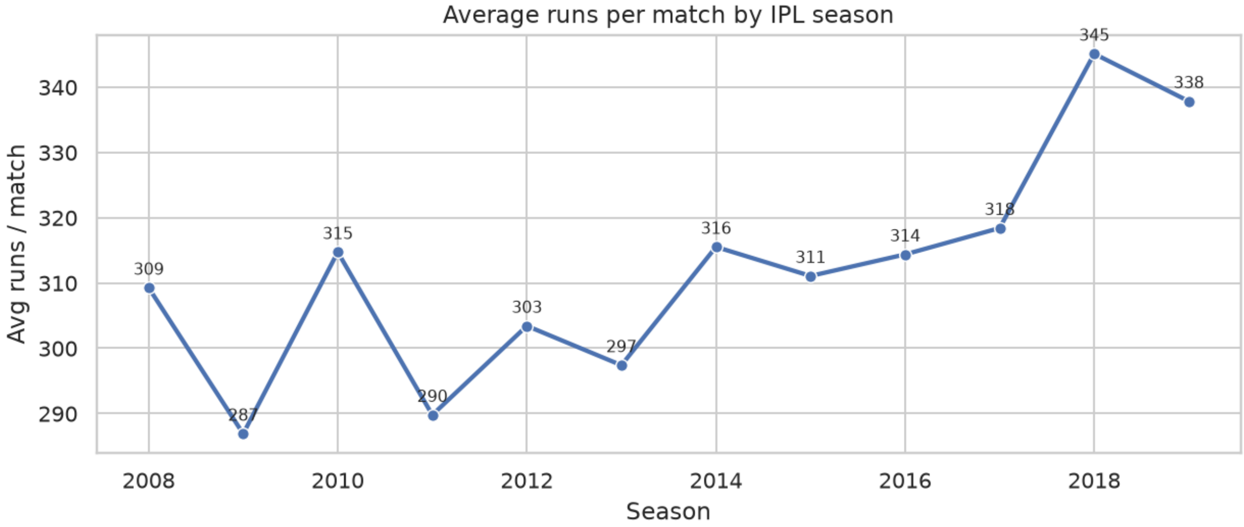

Fig 5. Average runs per match by season.

Scoring climbed from ~309 runs/match in 2008 to ~338 in 2019. The 2018 peak (~345) coincides with the return of CSK after a two-year ban and an Impact Player-friendly format.

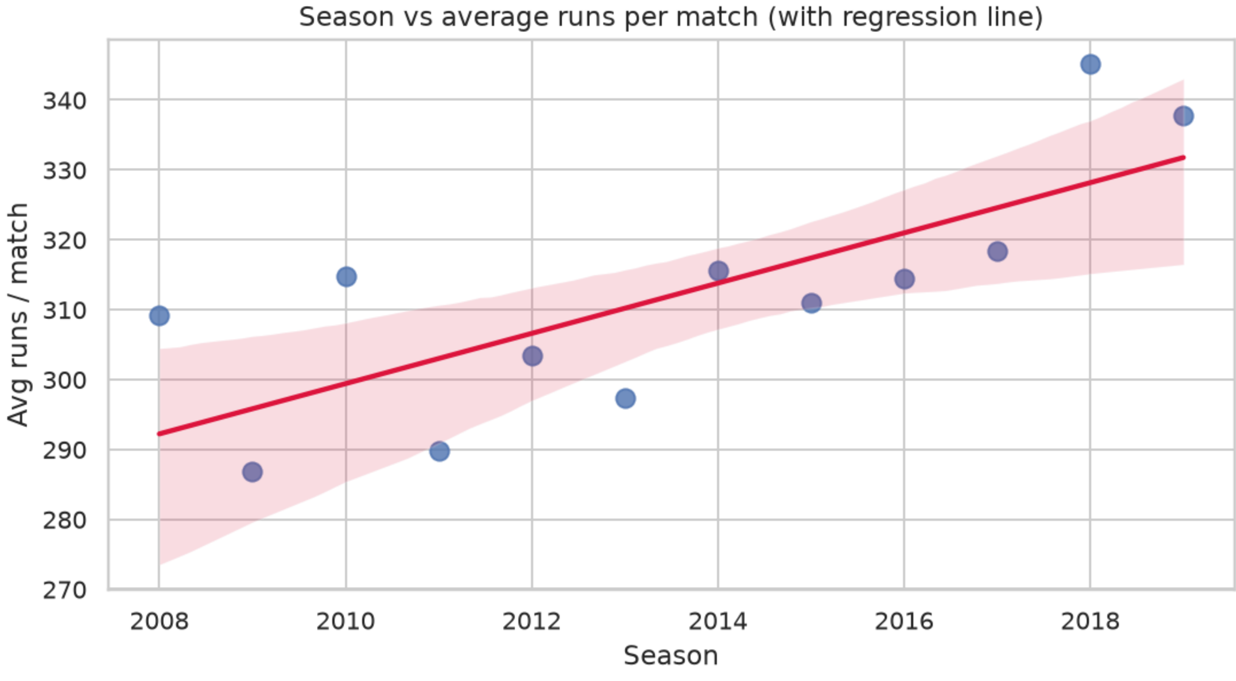

Fig 6. Regression line confirming an upward scoring trend.

OLS slope = +3.59 runs per season - yes, the IPL is becoming measurably more high-scoring.

3.4 Match Outcome & Venue Analysis

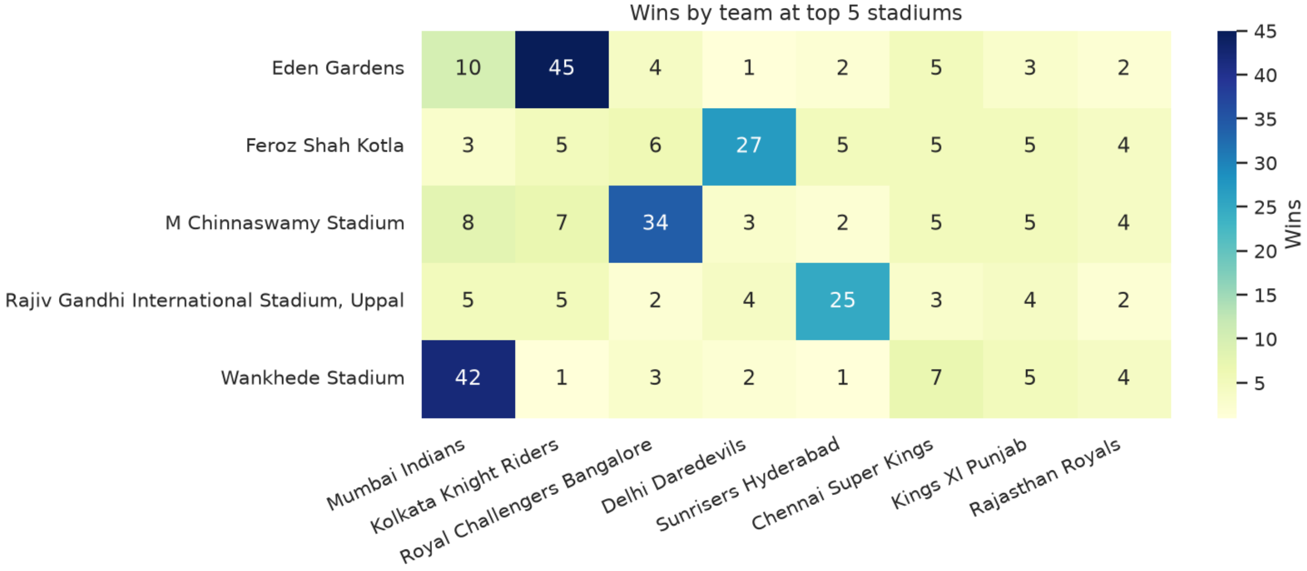

Fig 7. Wins by team at the top-5 stadiums (by matches hosted).

M Chinnaswamy Stadium and Wankhede Stadium show the strongest 'home' effect - RCB and MI win disproportionately at their home grounds. Eden Gardens (KKR) and Feroz Shah Kotla (DD) show weaker but still positive home tilts.

3.5 Death-Over Performance (overs 16-20)

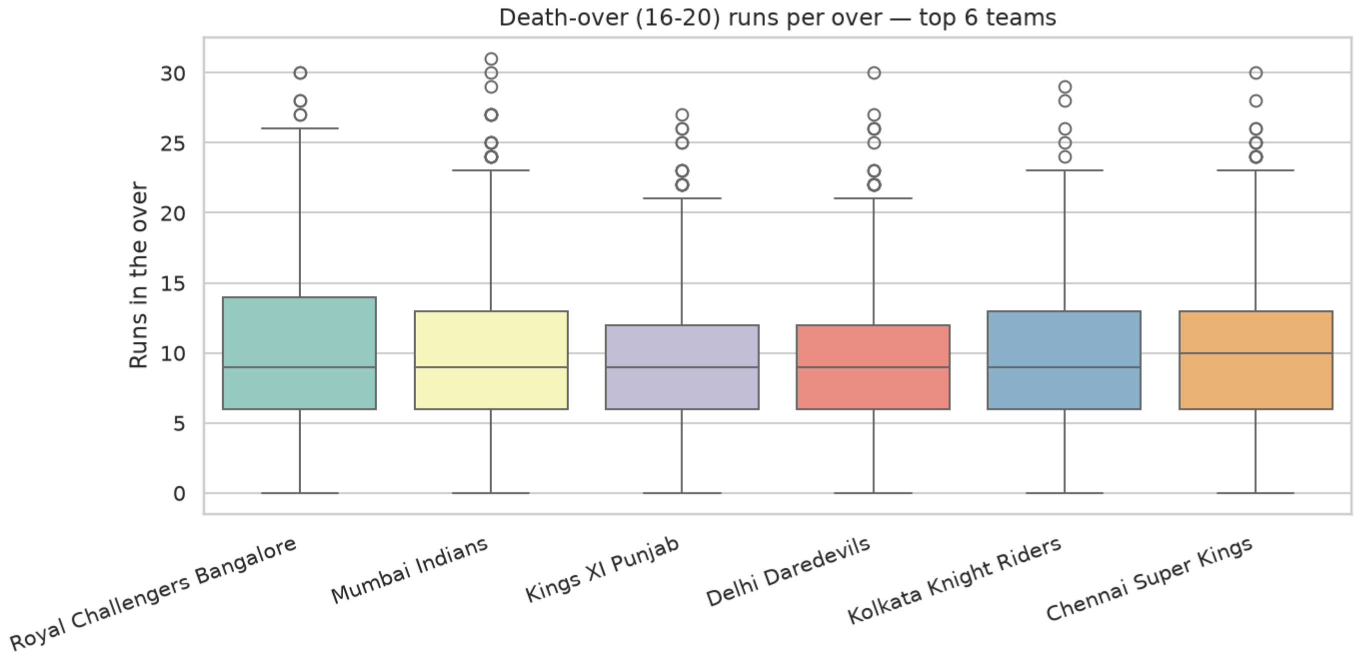

Fig 8. Runs per death over distribution for the top 6 teams.

CSK posts the highest median (10) at the lowest variance (std 5.14) - the most CONSISTENT finishers. RCB matches CSK on mean (10.33) but with the largest spread (std 5.57) - the most EXPLOSIVE-BUT-VOLATILE side at the death.

4. Recommendation for Team Strategy

Single, actionable recommendation:

"On winning the toss at a high-scoring or dew-prone venue (Chinnaswamy, Wankhede, Uppal), elect to FIELD first."

Rationale: across 12 seasons the field-first option converts the toss into a win ~56 % of the time versus ~46 % for bat-first, and the gap widens at venues where average scores are already high. A captain who anticipates the target gets to (a) calibrate batting tempo precisely, (b) use bowling changes to defend specific milestones, and (c) exploit dew in the second innings. Bat-first should be reserved for slow tracks (e.g. Chepauk) and day matches where dew is irrelevant.

5. Self-Reflection & Dataset Limitations

- The dataset stops at 2019, so the COVID-era 'bubble' seasons and the current 10-team / Impact-Player format are not represented.
- There is no pitch-report, weather or dew data, which limits how deeply we can explain WHY chasing dominates (dew vs pitch wear vs psychology).
- Ball-tracking data (line, length, speed, trajectory) is absent - we can see WHAT dismisses a batter (caught/bowled) but not WHERE the ball was.
- Player-level metadata (age, role, salary) would let us test if recent auction spend correlates with on-field win share.
- Limitation in our own analysis: economy-rate uses legal balls only; an extension would compute strike-rate, dot-ball % and phase-wise economy.

Given more time, the next iteration would (i) extend the dataset through 2024 via Cricsheet, (ii) join ball-tracking data, and (iii) build a logistic-regression win-probability model that updates after every over - useful for both commentary and in-game strategy.